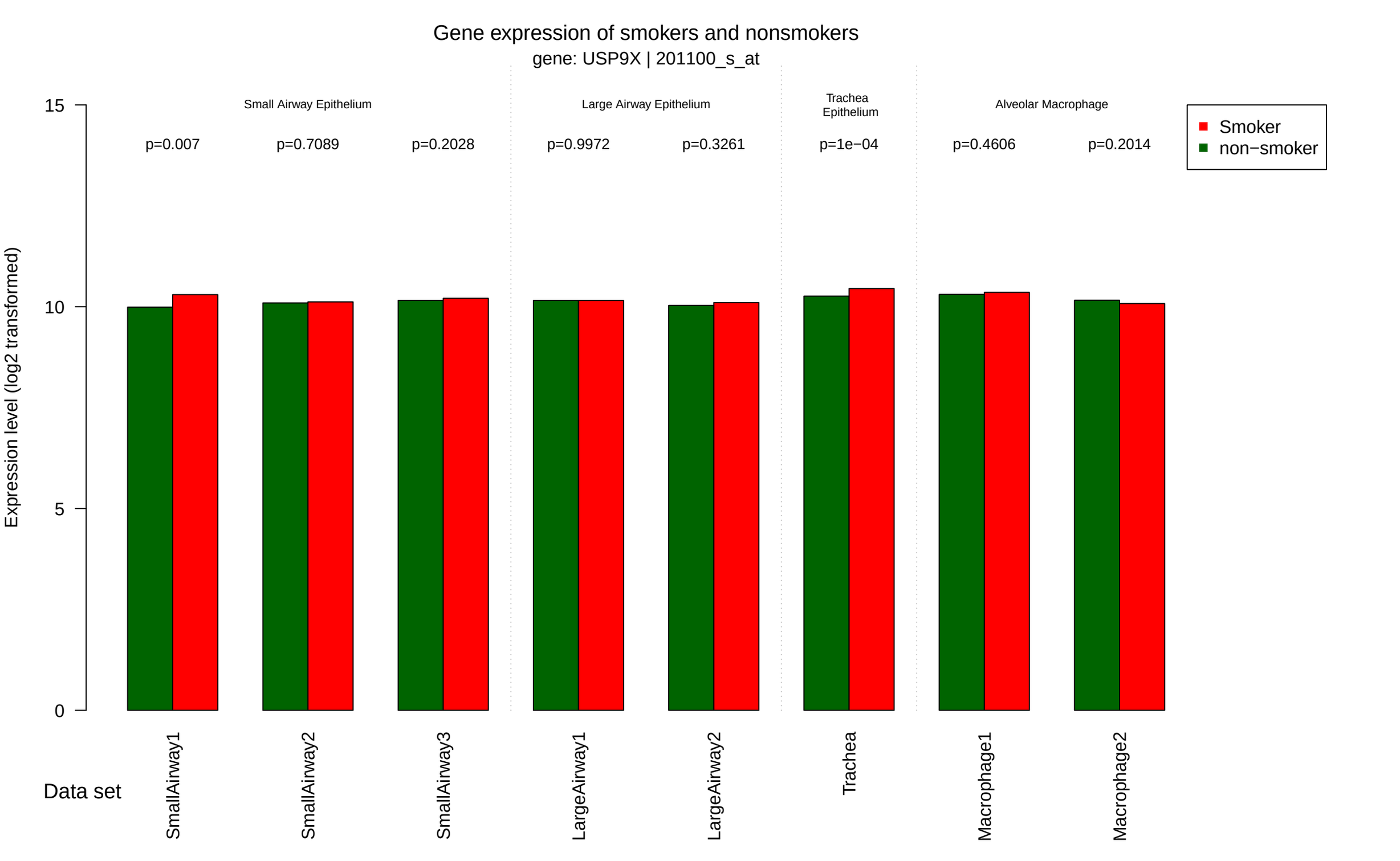
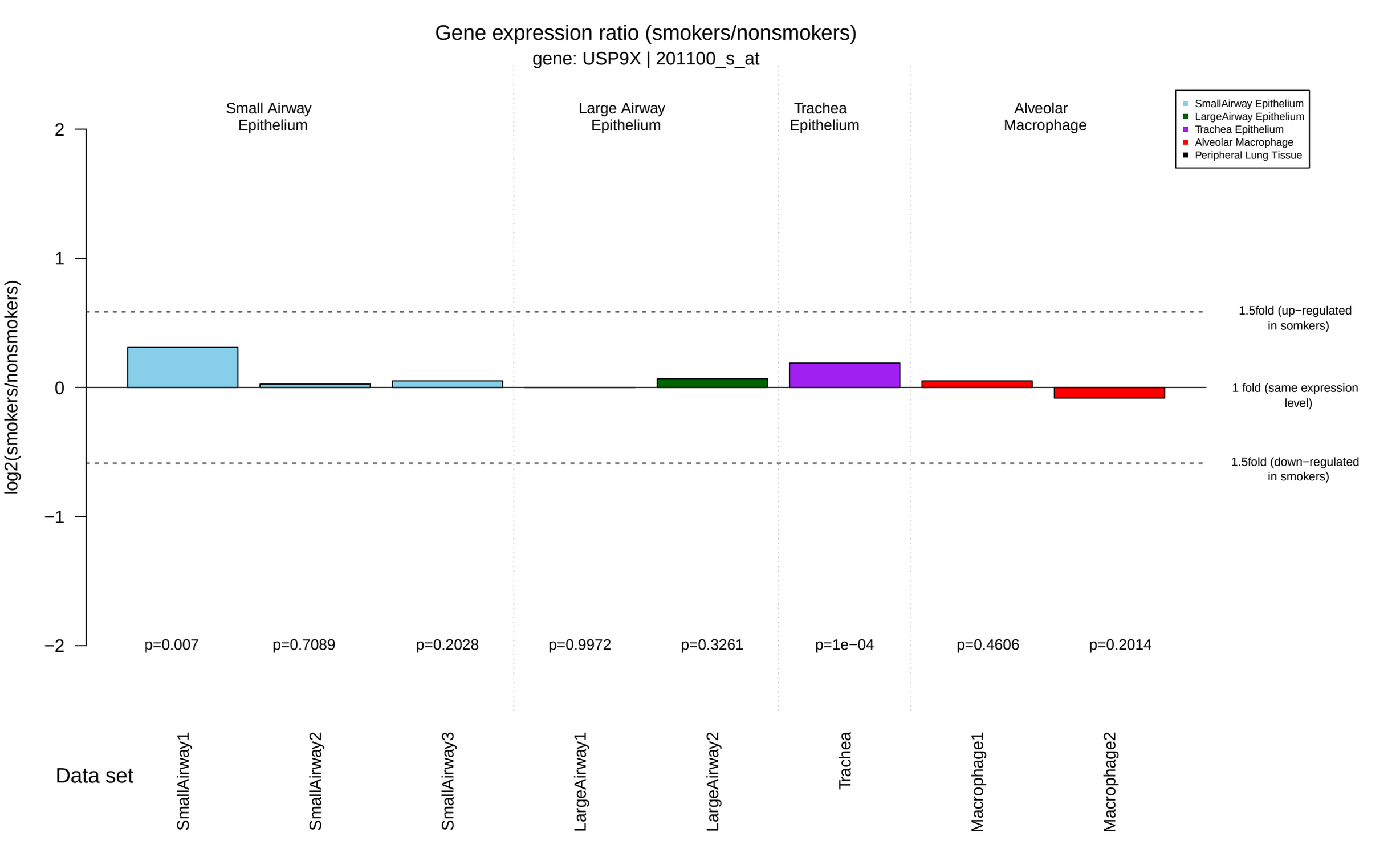
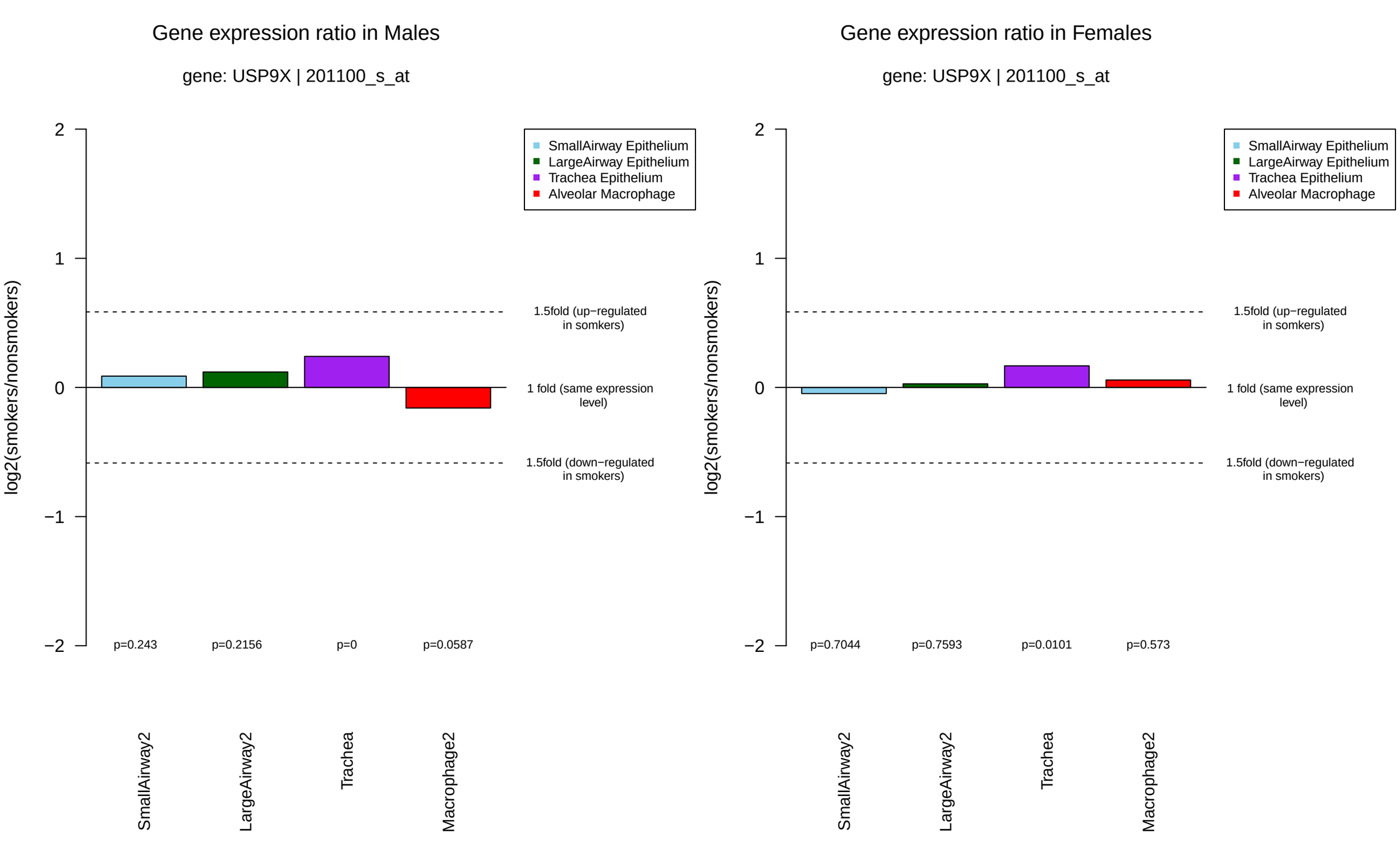
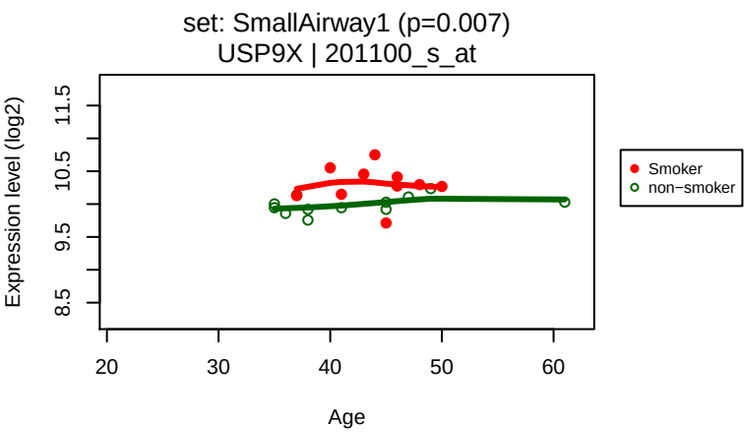
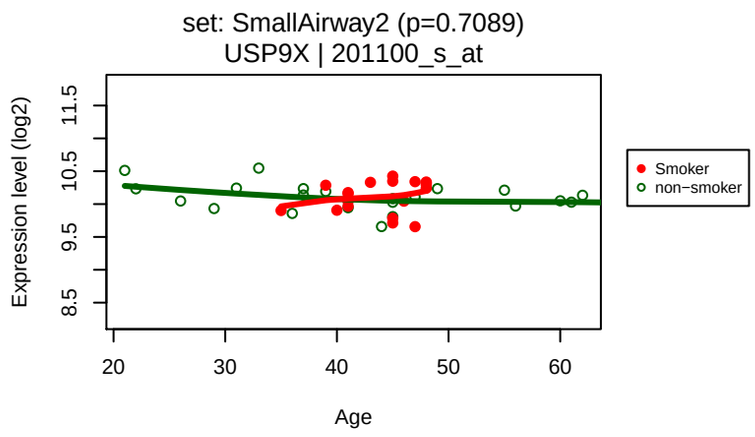
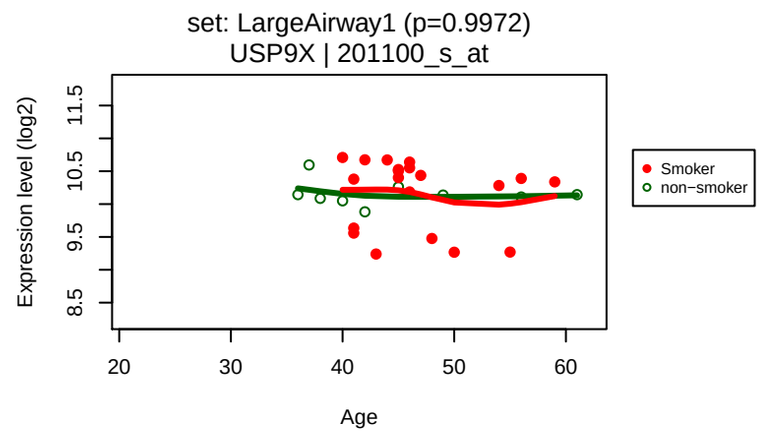

Gene expression of smokers and nonsmokers
gene: USP9X | 201100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.007
p=0.7089
p=0.2028
p=0.9972
p=0.3261
p=1e−04
p=0.4606
p=0.2014
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

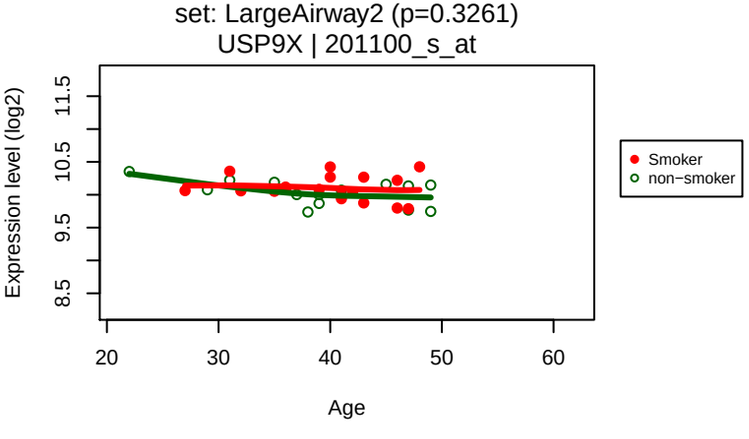

Gene expression ratio (smokers/nonsmokers)
gene: USP9X | 201100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.007
p=0.7089
p=0.2028
p=0.9972
p=0.3261
p=1e−04
p=0.4606
p=0.2014
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

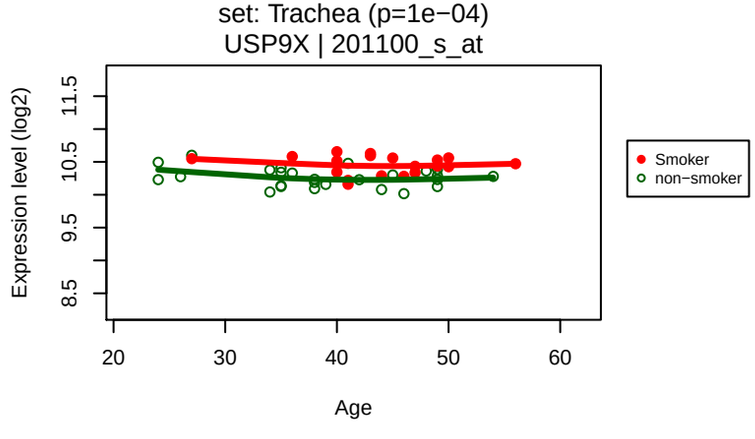

Gene expression ratio in Males
gene: USP9X | 201100_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.243
p=0.2156
p=0
p=0.0587
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP9X | 201100_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7044
p=0.7593
p=0.0101
p=0.573
SmallAirway2
LargeAirway2
Trachea
Macrophage2

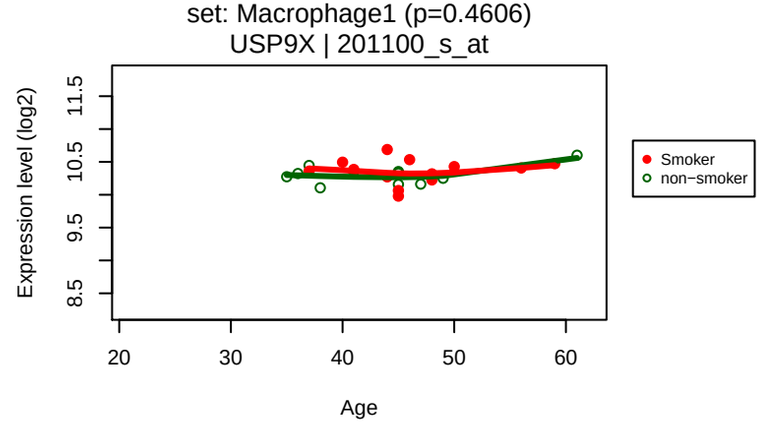

set: SmallAirway1 (p=0.007)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker

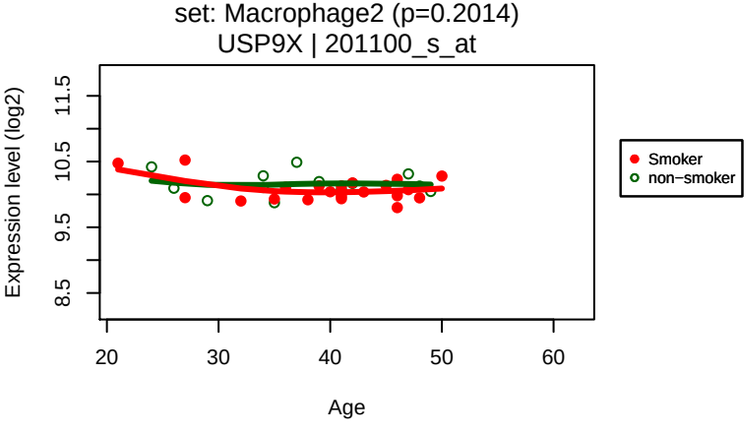

set: SmallAirway2 (p=0.7089)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker

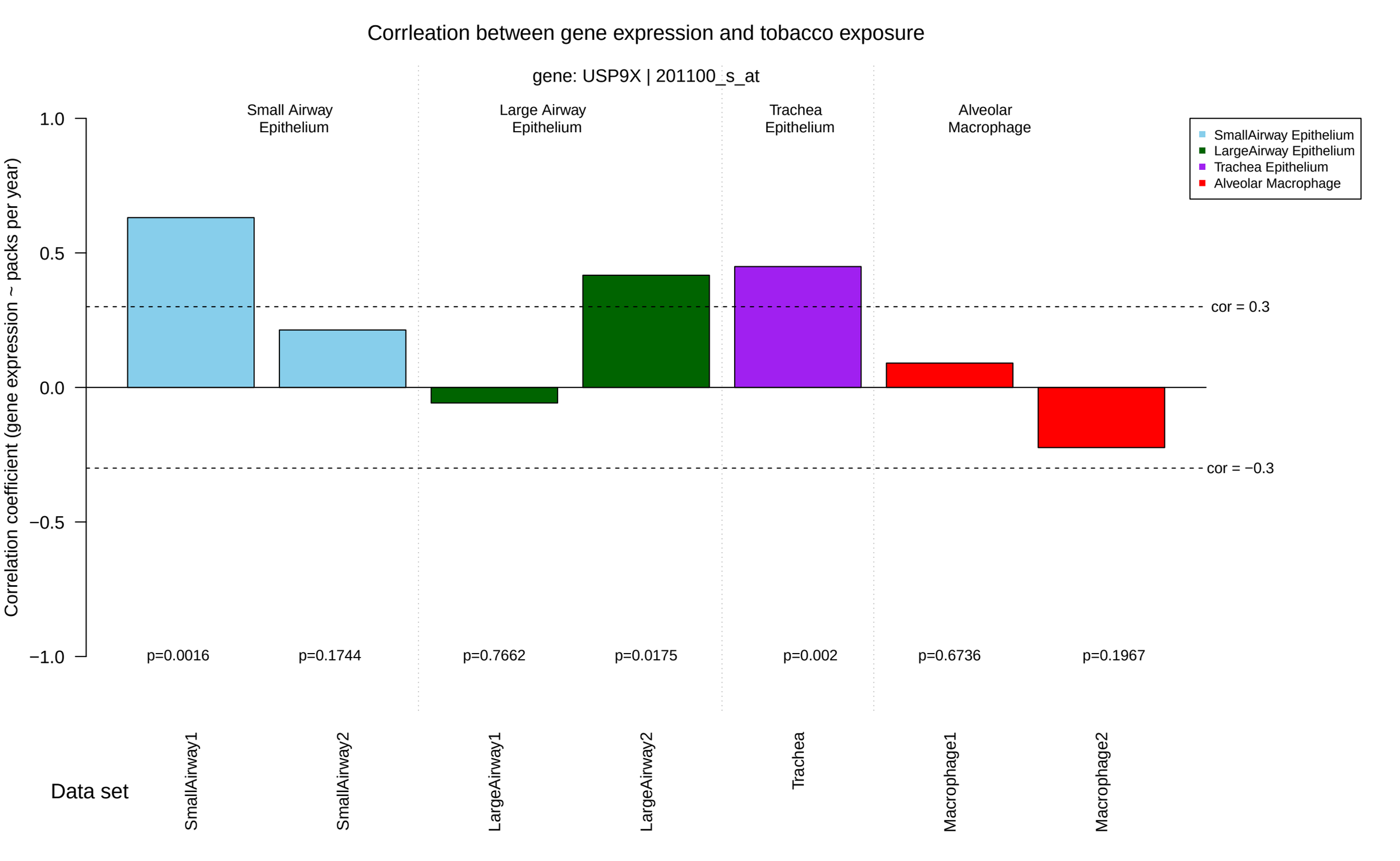

set: LargeAirway1 (p=0.9972)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3261)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4606)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2014)
USP9X | 201100_s_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP9X | 201100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0016
p=0.1744
p=0.7662
p=0.0175
p=0.002
p=0.6736
p=0.1967
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2